Web Application for DPCfam and DPCstruct Data Exploration

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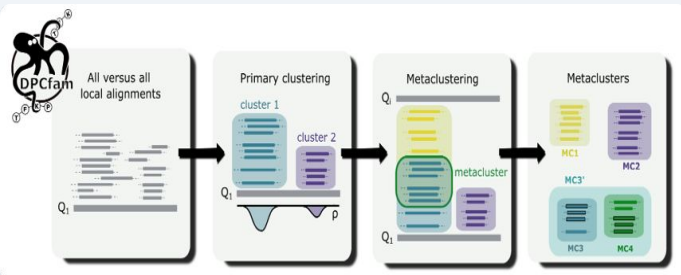
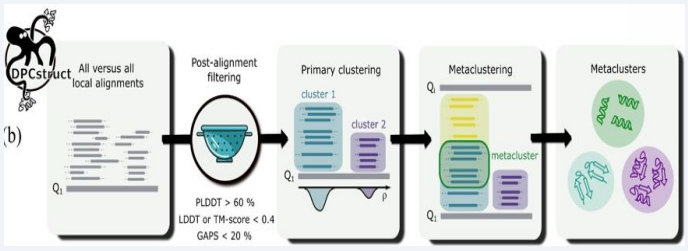
1 Motivation

- Proteins are the workhorses of every living cell; yet only a handful has been experimentally characterized.
- Protein domain families can support functional hypotheses for unannotated sequences.
- Manual curation improves annotation quality, but it does not scale with the rapid growth of protein databases.
- The Density Peak Clustering (DPC) algorithm provides fully unsupervised, automated classification of protein domains into families called ***metaclusters***.
- **DPCfam** [1]: 23 M UniRef50 sequences -> 45 K sequence-based metaclusters.

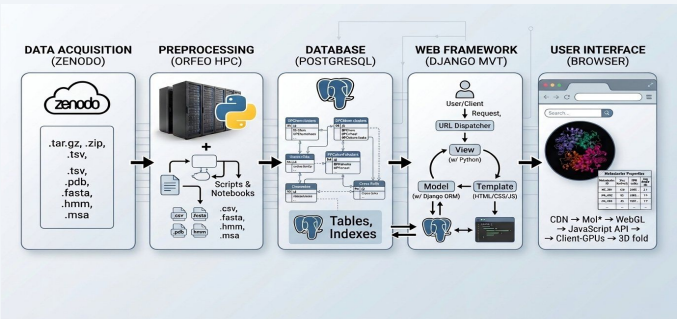


- **DPCstruct** [2]: 15 M AlphaFold2-predicted structures -> 28 K structure-based metaclusters.



- We develop a unified Django application to contribute to the **FAIR**ification of both datasets, enabling interactive exploration and downstream analysis.

2 Methodology

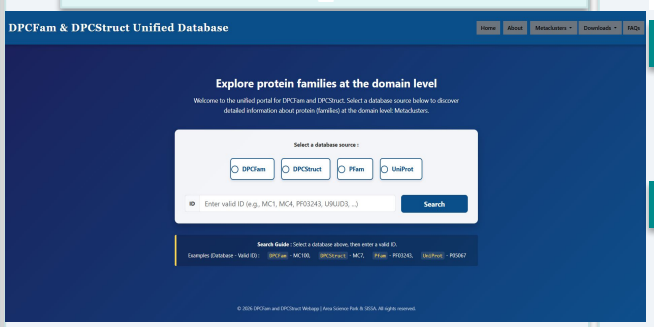


- Download DPCfam and DPCstruct datasets from Zenodo.
- Develop Python scripts and Jupyter notebooks for data exploration and preprocessing.
- Create and populate a PostgreSQL database with optimised tables and indexes.
- Build a full-stack Django application with a custom-coded HTML, CSS, and JavaScript frontend.
- Validate the user interface across representative query types.

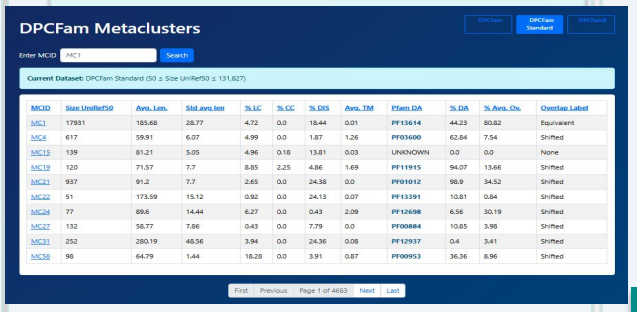
3 Results

The application supports four query types from the landing page : DPCfam Metacluster ID (MCID), DPCstruct Metacluster ID (MCID), Pfam ID (Familiy or Clan), and UniProt ID (Protein Identifier from UniProt).

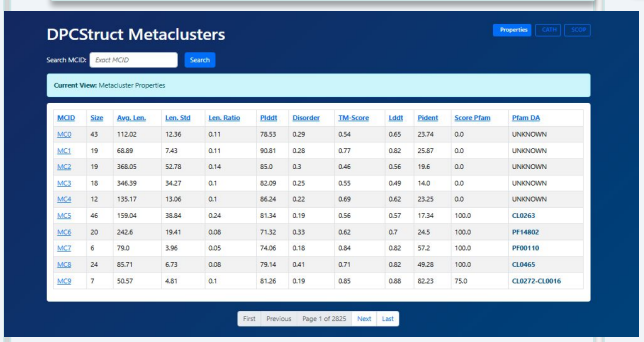
User Interface : Landing Page



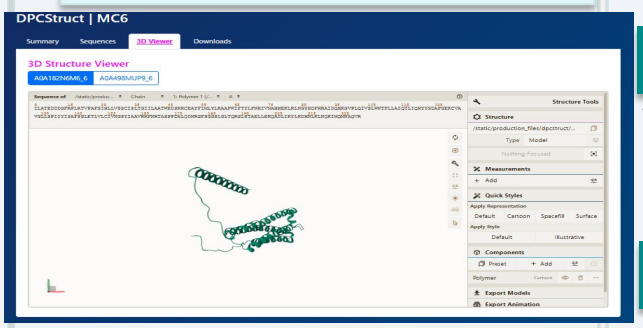
DPCfam Metaclusters



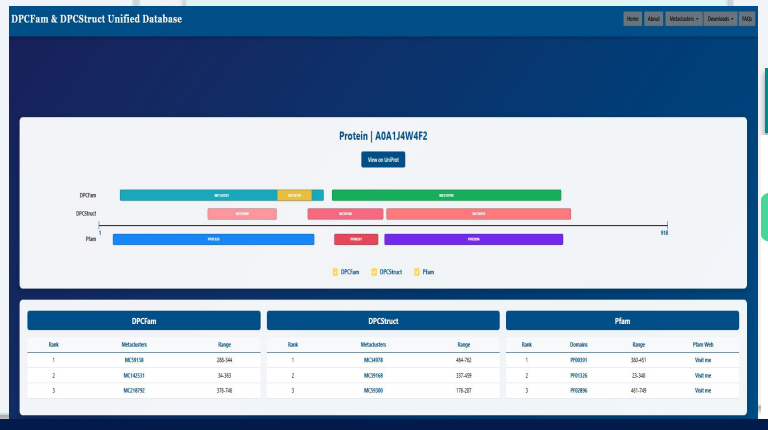
DPCstruct Metaclusters



3D Structure Viewer (Mol* Integration)



Matching samples per Protein ID



4 FAIR Data Principles

F - Findable

Metaclusters are identified by unique, persistent IDs and are accessible via structured queries.

A - Accessible

All data can be retrieved through a public web interface and without requiring authentication.

I - Interoperable

Standard biological identifiers (UniProt, Pfam) and open file formats (CSV, PDB, FASTA,…) are used throughout.

R - Reusable

Datasets are published on Zenodo with clear provenance metadata and permissive licences.

5 Conclusion

- A unified web application was developed to enable interactive exploration of DPCfam and DPCstruct metaclusters.
- Users can query by DPCFam MCID, DPCstruct MCID, Pfam ID, or UniProt ID and visualise representative 3D protein structures in the browser.
- Once polished, the tool will be deployed online to support downstream annotation analyses.
- Future work will expand query types and integrate functional annotation .

Acknowledgement

This work was carried out at the Laboratory of Data Engineering (LADE), Area Science Park, as part of a Professional Master's thesis in Data Management and Curation (MDMC) at SISSA .

References

- [1] Russo, G. et al. Dpcfam: unsupervised protein family classification by density peak clustering of large sequence datasets. (2022).
- [2] Barone, F. et al. Unsupervised domain classification of alphafold2-predicted protein structures. (2025).

Reproducibility - Contact

Scan Me

GitHub

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